discussed in Chapter 22, radioecologists the uptake of radioactive elements by plants and animals, as described in Chapter 15, etc.

A number of ideas have been proposed for the peaceful use of nuclear explosives. It has been estimated that the explosive power of nuclear charges is sufficiently great that their use for large scale explosions would be much cheaper than the use of conventional explosives. About 136 explosions have been used for civil engineering projects in the former USSR; sizes varied from <1 kt to ~50 kt. In the United States crater formation and gas stimulation experiments have been made. Suggested uses include the blasting of harbours on remote coasts to allow greater access to inland mineral deposits, crushing of ore bodies in order to obtain valuable minerals, crushing of underground rocks which may hinder the use of natural gas reservoirs, diverting asteroids threatening the earth, etc.

19.19. EXERCISES

19.1. If the energy developed by a 20 Mt fusion weapon could be used for producing electricity at a value of 2 cents kWh^{-1} what would the "electric value" of the device be? Such a weapon may be expected to cost $ 10 million. One ton TNT releases 1 Gcal of energy in an explosion.

19.2. The bomb over Hiroshima contained ^{235}U. How many grams were fissioned to correspond to 15 kt of TNT?

19.3. Compare two 500 MW_e electric power stations, one burning oil and the other using 3.0% enriched uranium. Both stations operate 6000 h y^{-1} at 35% efficiency (heat to electricity). The oil (43.5 MJ/kg combustion energy) is carried by 100 000 ton d.w. (dead weight, i.e. carrying capacity) oil tankers, and the uranium fuel by train cars of 20 t capacity each. (a) How many oil tankers will be needed every year for the oil-fired station? How many train cars will be needed every year for the nuclear power station for transporting (b) the enriched UO_2 reactor fuel, (c) the corresponding amount of natural uranium as U_3O_8 to the isotope enrichment plant, if the tail is 0.35% in ^{235}U? (See Ch. 3 and §19.9.) Reactor fuel rating 40 $MW_{th}d/kg$ U.

19.4. Calculate the number of collisions required to reduce a fast fission neutron ($\bar{E}$ = 2 MeV) to thermal energy (E_n 0.025 eV) in a light-water-moderated reactor, assuming that the data in Table 19.3 are valid.

19.5. Calculate the thermal fission factor for a mixture of 60% ^{239}Pu, 30% ^{240}Pu, and 10% ^{241}Pu.

19.6. The world's first water boiler reactor (LOPO, Los Alamos, 1944) was a homogeneous solution of enriched uranium sulfate as follows: 580 g ^{235}U, 3378 g ^{238}U, 534 g S, 14068 g O, and 1573 g H. From these values, and Tables 19.2 and 19.3, calculate η and f; neglect S. With $p = 0.957$, what will k_∞ be?

19.7. A large homogeneous thermal reactor contains only ^{235}U dispersed in beryllium in the atomic ratio $1{:}3 \times 10^4$. The migration area is $0.023\ m^2$. Assuming $p = \varepsilon = 1$, calculate the size of a cylindrical reactor with height equal to diameter.

19.8. A cubic unreflected graphite moderated natural uranium reactor contains 3% enriched uranium as UC homogeneously dispersed in the graphite matrix; the weight ratio $C/U = 10$. The resonance passage and thermal utilization factors are both assumed to be 0.9; $\varepsilon = 1.00$. Make an estimate of the critical size of the cube.

19.9. The LOPO reactor in exercise 19.6 has a neutron age $\tau = 31.4\ cm^2$, and diffusion area L^2 $1.87\ cm^2$. Calculate (a) the fast neutron leakage factor, and (b) the critical radius for the homogeneous sphere, if $k_\infty = 1.50$.

19.10. Our solar system is considered to be 4.5 billion years old. What was the ^{235}U percentage in natural uranium when the solar system was formed?

19.11. The radiometric sensitivities for discovering ^{59}Fe, ^{131}I, and ^{90}Sr are 75, 25, and $0.74\ kBq\ m^{-3}$ of water. In the Würgassen plant the total permitted aqueous annual release was 17 Ci β-emitters. Assume an activity ratio in the cooling water of 100:10:1 for the three nuclides above and that none of these activities exceed 1% of the permissible release. How many times must a liquid sample taken each day be concentrated to meet these requirements?

19.12. What amount of tritium (Bq) is produced in the Würgassen nuclear plant assuming that ^3H is only produced through capture in the deuterons of the original cooling water, the amount or which is 50% of the core volume? Data on fluxes, cross sections, and releases are given in Table 19.4. Neglect the tritium decay rate.

19.13. A BWR has operated at full power for a week. At what time after a scram would the Xe poisoning reach its maximum? Use data in Table 19.4.

19.14. The net efficiency of a 1000 MW_e BWR is 34.1%. Estimate the amount of ^{235}U consumed during its first day of full power operation with a completely fresh load of fuel.

19.15. Consider a power reactor in which microspheres ($r = 0.3$ mm) of frozen 1:1 T-D mixture (density $170\ kg\ m^{-3}$) are fused by laser irradiation. The laser compresses the spheres to $N_v = N_v^o 10^4$, where N_v^o is the number of atoms per m^3 at ordinary pressure, and also heats it to a temperature corresponding to almost 20 keV. The energy developed through the T-D fusion reaction leads to expansion of the spheres, which occurs with the velocity of sound ($v_s = 10^8\ m\ s^{-1}$). This leads to no more than 25% of the particles fusing. What power is produced if the fusion micro-explosions occur at a rate of 30 per s?

19.16. Deuterium is to be injected into a fusion reactor at a density of $10^{20}\ D^+$ and $10^{20}\ e^-\ m^{-3}$ and at an energy of 100 keV. How much of the deuterium must fuse to compensate for the ionization and injection energy? The ionization energy of the deuterium atom is 13 eV.

19.20. LITERATURE

H. D. Smyth, *Atomic Energy for Military Purposes*, Princeton University Press, 1946.
S. Glasstone (Ed.), *The Effects of Nuclear Weapons*, USAEC, Washington, 1962.
J. K. Dawson and R. G. Sowden, *Chemical Aspects of Nuclear Reactors*, Butterworths, 1963.
S. Peterson and R. G. Wymer, *Chemistry in Reactor Technology*, Pergamon Press, Oxford, 1963.
ANL-5800, *Reactor Physics Constants*, USAEC, TID, 1963.
Reactor Handbook, 2nd edn., Interscience, 1960–1964.
J. R. Lamarch, *Introduction to Nuclear Reactor Theory*, Addison-Wesley, 1966.
J. G. Wills, *Nuclear Power Plant Technology*, Wiley, 1967.
C. O. Smith, *Nuclear Reactor Materials*, Addison-Wesley, 1967.
E. Teller, W. K . Talley, G. H. Higgins, and G. W. Johnson, *The Constructive Uses of Nuclear Explosives*, McGraw-Hill, 1968.
G. I. Bell and S. Glasstone, *Nuclear Reactor Theory*, van Nostrand, 1970.
W. Fratzscher and H. Felke, *Einführung in die Kerntechnik*, VEB Deutscher Verlag fur Grundstoffindustrie, Leipzig, 1971.
U.N. Conferences on peaceful uses of atomic energy, 1955, 1958, 1965, 1972.
J. A. Richardson, Radioactive waste quantities produced by light water reactors, *Nucl. Eng. Int.* Jan. (1974) 31.
J. R. Lamarch, *Introduction to Nuclear Engineering*, Addison-Wesley, 1975.
IAEA, *The Oklo Phenomenon*, Vienna, 1975.
G. A. Cowan, A natural fission reactor, *Sci. Am.* July 1976.
IAEA, *Management of Radioactive Wastes from the Nuclear Fuel Cycle*, Vols. 1 and 2, Vienna, 1976.
Jahrbuch der Atomwirtschaft 1977, Handelsblatt GmbH, Verlag fur Wirtschaftsinformation, Dusseldorf, Frankfurt, 1977.
Int. Conf. on Nuclear Power and its Fuel Cycle, Salzburg 1977, IAEA, Vienna, 1977.
R. A. Rydin, *Nuclear Reactor Theory and Design*, PBS Publications, 1977.
H. Edelhäuser, H. Bonke, I. Gans, O. Huber, K. J. Vogt, and R. Wolter, *Radioactive Effluents and Present and Future Radiation Exposure to the Population from Nuclear Facilities in the Federal Republic of Germany*, in Int. Conf. on Nuclear Power and its Fuel Cycle, Salzburg 1977, IAEA, Vienna, 1977.
H. W. Graves, Jr., *Nuclear Fuel Management*, J. Wiley & Sons, 1979.
M. Benedict, T. H. Pigford and H. W. Levi, *Nuclear Chemical Engineering*, 2nd Ed., McGraw-Hill, 1981.
The Management of Radioactive Waste, The Uranium Institute, London, 1991.
World Nuclear Industry Handbook – 2012, Vol. 57, No 697, Nuclear Engineering International, ISSN 0029-5507, Aug. 2012.

Nuclear explosives

Nuclear Weapons. Report of the Secretaty-General of the United Nations, Autumn Press, 1981.
S. Glasstone and A. Sesonske, *Nuclear Reactor Engineering*, Van Nostrand Reinhold, 3rd Ed., 1981.
W. Seifritz, *Nucleare Sprengkörper*, Karl Themig, 1984.
U. Ivarsson (Ed.), *FOA orienterar om kärnvapen* (in Swedish), the Swedish National Defense Research Institute, **15** (1990).

Controlled fusion

M. J. Monsler, J. Hovingh, D. L. Cook, T. G. Frank and G. A. Moses, An overview of inertial fusion reactor design, *Nucl. Techn. Fusion*, **1**, July 1981, 302.
J. R. Roth, The Impact of Engineering Constraints on the Feasibility of Advanced Fuel Fusion Reactors, *Nucl. Techn. Fusion*, **2** (1982) 29.
P. A. Ertaud, Fusion thermonucleaire controlee – la grande illusion, *RGN*, Mai-Juin 1985, 242.
R. W. Conn, V. A. Chuyanov, N. Inoue and D. R. Sweetman, The International Thermonuclear Experimental reactor, *Sci. Am.*, April 1992, 75.
J. G. Cordey, R. J. Goldston and R. R. Parker, Progress Toward a Tokamak Fusion Reactor, *Physics Today*, Jan. 1992, 22.
ITER-project link: http://www.f4e.europa.eu/